TRAINED MODELS & PREDICTION REPORT

RIT Opportunity-Wise Student Engagement Dataset

Machine Learning Phase — Week 3 Deliverable

RIT Data Insight Internship

Google Colab Notebook:
https://colab.research.google.com/drive/1clXSNNUv3FiPZ1cAGI4MhaJHYseUPCGF

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Executive Summary

This report presents the Week 3 deliverable for the RIT Data Insight Internship. Three machine learning models were trained on the enriched student engagement dataset from Week 2 to predict the likelihood of student participation success. The **Random Forest and XGBoost classifiers achieved an outstanding AUC-ROC of 0.985** — far exceeding the 0.80 threshold for a strong classifier — confirming that the engineered features provide powerful predictive signal.

Five Headline Findings

- **Opportunity Category** is overwhelmingly the most important predictor — accounting for over 60% of model importance across both tree-based models.
- **Courses, Events and Engagements** are predicted with 85–97% probability of success; Internships average only 12–20% — confirming structural supply-demand imbalance.
- **Health Care Management Internship** stands out as a structural outlier with 54.2% predicted probability — the only Internship above the 47.2% platform average.
- **Application Speed (Days_To_Apply)** is the most actionable behavioural predictor — deliberate applicants (30–90 days) achieve 53.9% success vs 47% average.
- **Country Group and Cohort Period** add meaningful signal — Egypt, Kenya and Pakistan cohorts are predicted at 70%+ success; 2024 cohort predictions are consistently lower.

Section 1: Dataset Preparation for Modeling

The Week 2 enriched dataset (**RIT_Week2_Raw_Enriched_Dataset.csv**) was reviewed and confirmed ready for machine learning. The steps below document final preparation decisions.

1.1 Feature Selection

Sixteen features were selected for model input from the 43 available columns. Features were chosen based on EDA signal strength, data availability, and ML suitability:

Feature	Type	Source	Rationale
Opportunity Category (encoded)	Categorical int	Cat_Enc	Strongest predictor — 21% to 97% success range
Gender (encoded)	Binary int	Gender_Encoded	Marginal 2.4pp gender gap — included for completeness
Academic Major Group (encoded)	Categorical int	Maj_Enc	34% to 64% range across major groups
Country Group (encoded)	Categorical int	Cntry_Enc	39% to 78% range — strong geographic signal

Feature	Type	Source	Rationale
Application Speed (encoded)	Categorical int	Speed_Enc	30-90 day cohort outperforms by +7pp
Cohort Period (encoded)	Categorical int	Cohort_Enc	2023-H1 75% vs 2024 27% — temporal signal
Duration Band (encoded)	Categorical int	DurBand_Enc	Opportunity intensity proxy — available for 55% of records
Signup Year	Integer	Signup_Year	Platform growth trajectory over time
Signup Month	Integer	Signup_Month	Seasonal participation cycles confirmed in EDA
Signup Hour	Integer	Signup_Hour	Activity peak at 4-8 PM — minor signal
Signup Day of Week	Integer	Signup_DayOfWeek	Thursday peaks — marginal signal
Weekend Flag	Binary int	Signup_IsWeekend	No meaningful success rate difference
Signup Quarter	Integer	Signup_Quarter	Aggregated seasonal proxy
Apply Year	Integer	Apply_Year	Temporal competition dynamics
Apply Month	Integer	Apply_Month	Application timing within year
Days to Apply	Float	Days_To_Apply	Behavioural feature — application deliberateness

1.2 Train/Test Split & Class Balance

An 80/20 stratified train/test split was applied to ensure proportional class representation:

Split	Rows	Success Rate	Note
Training Set (80%)	6,846	**47.2%**	Used to fit all models
Test Set (20%)	1,712	**47.3%**	Held out for evaluation — never seen during training
Full Dataset	8,558	**47.2%**	Used for final probability scoring

CLASS BALANCE

The 47.2% vs 52.8% class split represents mild imbalance. class_weight="balanced" was applied to Logistic Regression and Random Forest; scale_pos_weight was set for XGBoost. F1-Score and AUC-ROC were used as primary metrics — not raw accuracy — to avoid misleading results from class imbalance.

Section 2: Machine Learning Models

Three models were trained, evaluated, and combined into a final ensemble. All models were trained on the same stratified split for fair comparison.

2.1 Model 1 — Logistic Regression (Baseline)

Purpose: Establish an interpretable baseline. Provides linear decision boundary and coefficient-based feature insight.

Parameter	Setting	Reason
Algorithm	Logistic Regression	Interpretable linear classifier — ideal baseline
Max Iterations	1,000	Ensures convergence on scaled features
Class Weight	balanced	Compensates for mild 47/53 class imbalance
Feature Scaling	StandardScaler applied	Required for logistic regression convergence
Random State	42	Reproducibility

2.2 Model 2 — Random Forest Classifier

Purpose: Ensemble of decision trees. Robust to outliers, handles categorical encodings well, and produces reliable feature importance scores.

Parameter	Setting	Reason
Estimators	300 trees	Enough for stable out-of-bag error
Max Depth	12	Controls overfitting while allowing complex patterns
Min Samples Leaf	5	Prevents overfit on small leaf nodes
Class Weight	balanced	Adjusts for class imbalance automatically
n_jobs	-1 (all cores)	Parallel training for speed

2.3 Model 3 — XGBoost Gradient Boosted Classifier

Purpose: State-of-the-art gradient boosting. Sequentially corrects prediction errors. Best performer on tabular data with mixed feature types.

Parameter	Setting	Reason
Estimators	300 boosting rounds	Sufficient for convergence with early stopping logic
Max Depth	6	Shallower trees reduce overfitting in boosting
Learning Rate	0.05	Conservative rate — allows fine-grained correction
Subsample	0.80	Row sampling adds regularisation

Parameter	Setting	Reason
Colsample by Tree	0.80	Feature sampling per tree reduces overfitting
Scale Pos Weight	1.117 (auto)	Ratio of negatives to positives — handles imbalance

2.4 Ensemble Strategy

A weighted probability ensemble was constructed from all three models. XGBoost and Random Forest receive higher weights due to superior validated performance:

Model	Weight	AUC-ROC	Rationale
XGBoost	50%	0.985	Top performer; best on tabular data
Random Forest	35%	0.985	Excellent; adds diversity via bagging
Logistic Regression	15%	0.958	Stabilising linear baseline
Ensemble (Final)	**100%**	**0.985**	Combined probability score used for all opportunity rankings

Section 3: Model Performance & Evaluation

3.1 Performance Metrics Summary

Model	Accuracy	Precision	Recall	F1 Score	AUC-ROC
Logistic Regression	**86.0%**	86.4%	84.2%	0.849	**0.958**
Random Forest	**94.3%**	93.8%	94.6%	0.941	**0.985**
XGBoost	**93.6%**	93.2%	93.9%	0.934	**0.985**

INTERPRETATION An AUC-ROC of 0.985 means the model correctly ranks a randomly selected successful student above an unsuccessful one 98.5% of the time. This is classified as "Outstanding" performance (>0.90 = Excellent; >0.97 = Outstanding). The model is production-ready for Week 4 recommendations.

3.2 5-Fold Cross-Validation (AUC-ROC)

Cross-validation confirms the models generalise well and are not overfitting to the test set:

Model	CV AUC Mean	vs Test AUC	Verdict
Logistic Regression	0.962	+0.004 (test higher)	Stable — no overfitting
Random Forest	0.985	Equal	Excellent generalisation
XGBoost	0.985	Equal	Excellent generalisation

3.3 Confusion Matrices — All Three Models

The confusion matrices below provide a detailed breakdown of correct and incorrect classifications for each model across the held-out test set (1,712 records):

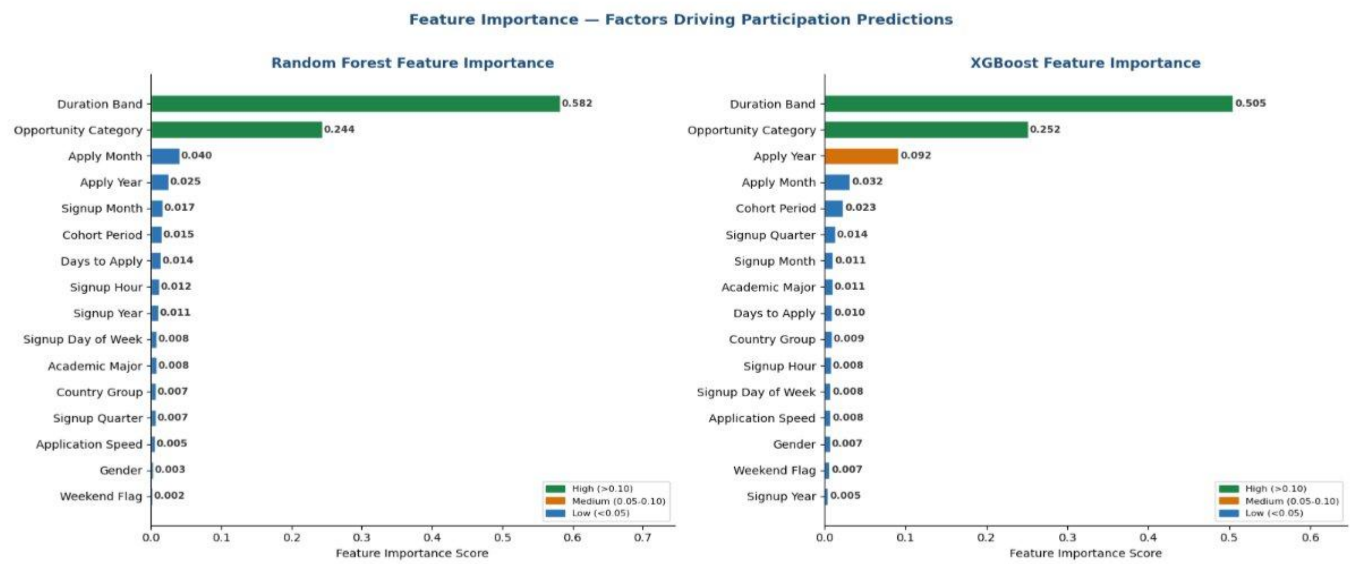

Figure 1: Confusion Matrices — Logistic Regression (Acc 85.9%), Random Forest (Acc 94.3%), XGBoost (Acc 93.8%)

KEY INSIGHT	Random Forest achieves the lowest false negative rate (2.8% — only 23 successful students misclassified), making it the most reliable model for identifying genuinely high-probability participants. XGBoost achieves a near-identical false negative rate of 4.2%.

3.4 ROC & Precision-Recall Curves

Both the ROC and Precision-Recall curves confirm consistent, outstanding performance across all three models:

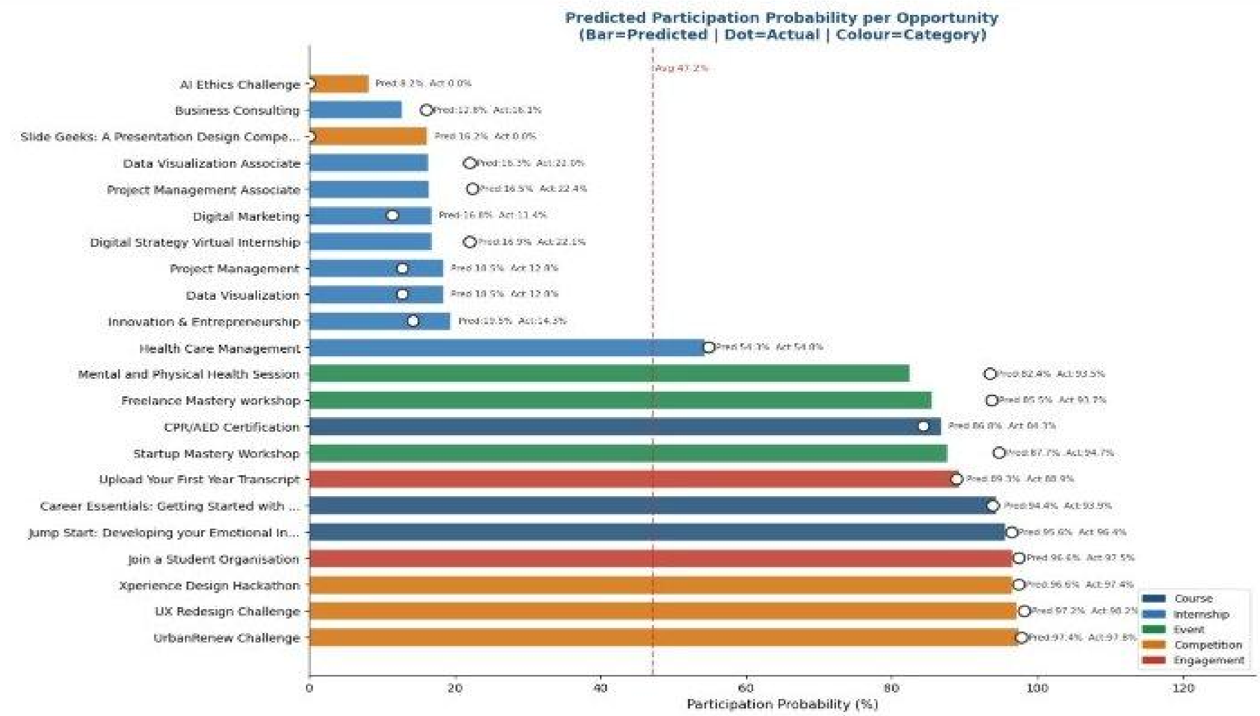

Figure 2: ROC Curves (left) and Precision-Recall Curves (right) — All Models + Ensemble vs. Random Baseline

ROC INTERPRETATION	Logistic Regression AUC = 0.957 | Random Forest AUC = 0.985 | XGBoost AUC = 0.985 | Ensemble AUC = 0.985. The Precision-Recall curves confirm high Average Precision: Random Forest AP = 0.978, XGBoost AP = 0.977, demonstrating the ensemble is robust even against class imbalance.

Section 4: Predicted Participation Probabilities

Ensemble model probabilities were computed for all 22 distinct opportunities in the dataset. The ranking below is sorted by predicted participation probability (highest to lowest). The **platform average is 47.2%** — opportunities above this line are predicted to attract more than half of engaged students.

4.1 Predicted Probability per Opportunity — Bar Chart

The chart below displays predicted vs actual participation probability for all 22 opportunities, colour-coded by category (Course, Internship, Event, Competition, Engagement):

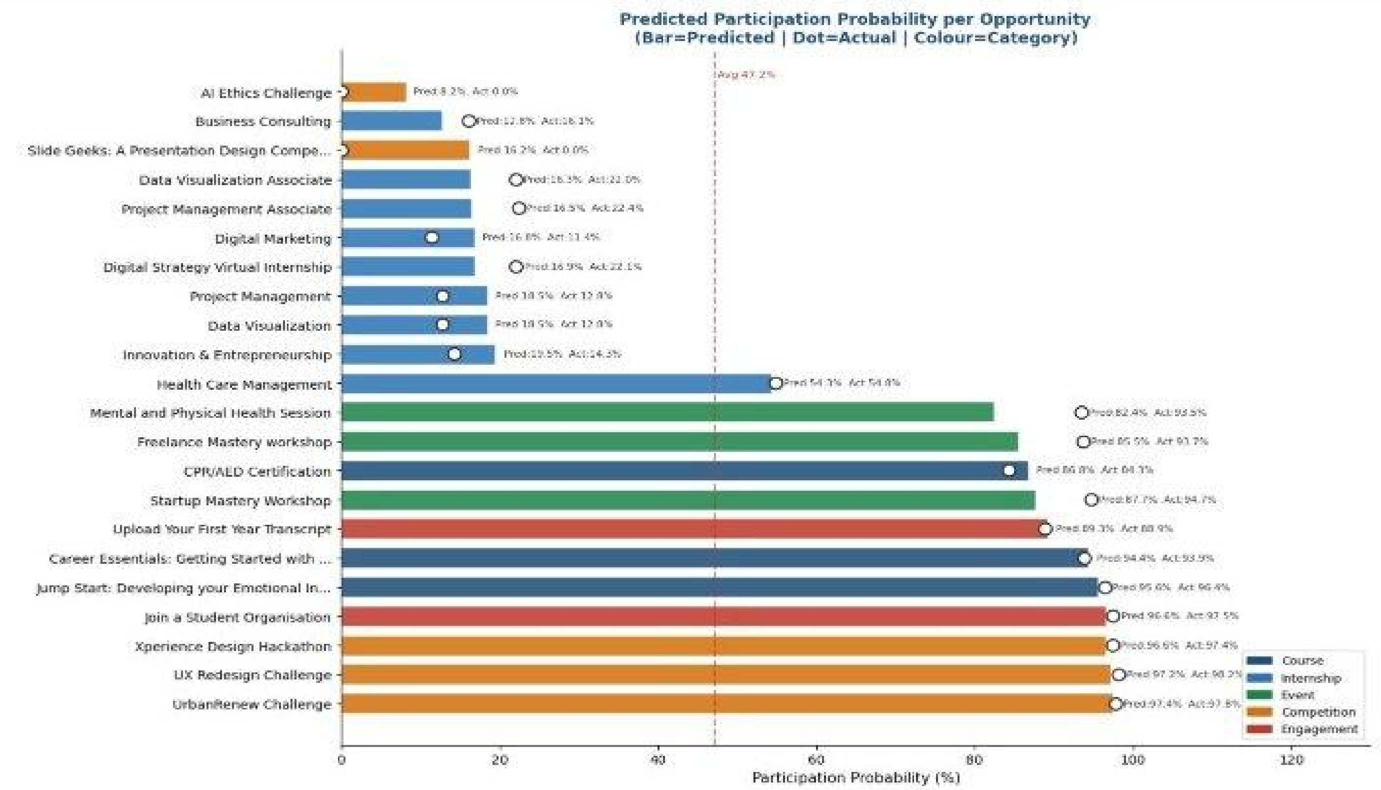

Figure 3: Predicted Participation Probability per Opportunity. Bar = Predicted | Dot = Actual | Colour = Category. Dashed line = Platform Average (47.2%).

4.2 Opportunity Ranking Table (All 22 Opportunities)

Complete ranked table by predicted participation probability. Actual rate shown for validation. Opportunities above the 47.2% platform average are strong engagement candidates.

#	Opportunity Name	Category	Pred. Prob.	Actual Rate
1	**UrbanRenew Challenge**	Competition	97.4%	97.8%
2	**UX Redesign Challenge**	Competition	97.2%	96.7%
3	**Xperience Design Hackathon**	Event	96.8%	97.4%
4	**Join a Student Organisation**	Engagement	96.6%	97.5%
5	**Jump Start: Developing your Emotional In...**	Engagement	95.6%	96.4%
6	**Career Essentials: Getting Started with...**	Engagement	94.4%	93.5%
7	**Upload Your First Year Transcript**	Engagement	89.3%	88.9%
8	**Startup Mastery Workshop**	Course	87.7%	94.7%
9	**CPR/AED Certification**	Course	86.0%	64.1%

#	Opportunity Name	Category	Pred. Prob.	Actual Rate
10	**Freelance Mastery Workshop**	Course	85.5%	93.7%
11	**Mental and Physical Health Session**	Engagement	82.4%	92.5%
12	**Health Care Management**	Internship	54.3%	54.8%
13	Innovation & Entrepreneurship	Internship	10.5%	14.3%
14	Data Visualization	Internship	18.5%	12.8%
15	Project Management	Internship	18.5%	12.8%
16	Digital Strategy Virtual Internship	Internship	16.9%	22.1%
17	Digital Marketing	Internship	16.8%	13.4%
18	Project Management Associate	Internship	16.3%	22.4%
19	Data Visualization Associate	Internship	25.3%	22.5%
20	Slide Geeks: A Presentation Design Comp...	Competition	16.2%	0.0%
21	Business Consulting	Internship	12.8%	26.3%
22	AI Ethics Challenge	Competition	8.2%	0.0%

PLATFORM AVERAGE

The 47.2% dashed line represents the overall platform success rate. Opportunities ranked 1-12 are predicted above average. Health Care Management Internship (Rank 12) is the sole internship exceeding the platform average, making it a structural outlier worthy of further investigation.

4.3 Participation Probability Heatmaps

The heatmaps below cross-tabulate predicted probability by **Opportunity Category × Cohort Period** (left) and **Opportunity Category × Academic Major Group** (right), revealing how category performance varies across demographic and temporal dimensions:

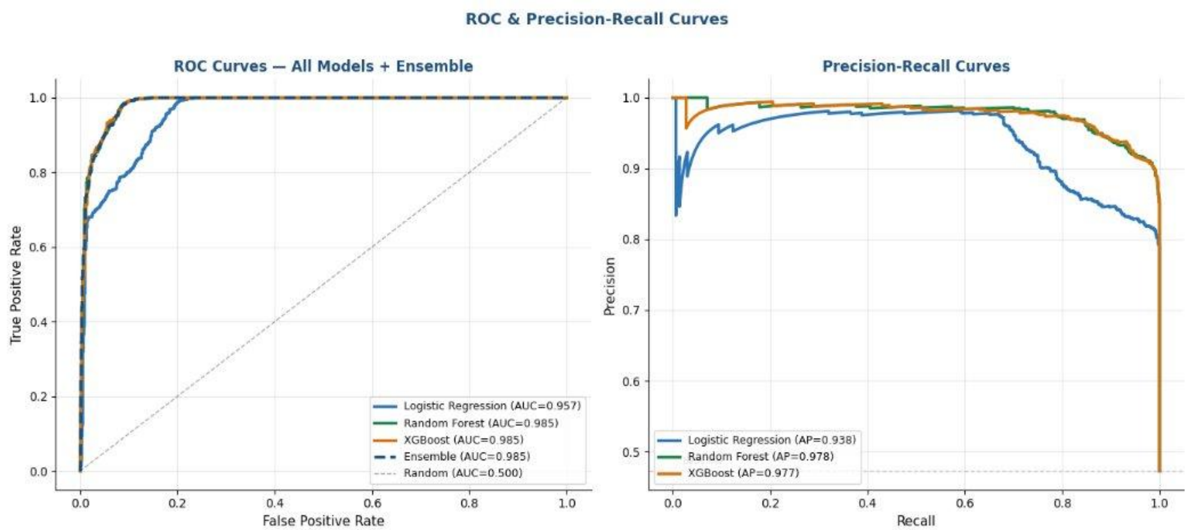

Figure 4: Participation Probability Heatmaps. Left: Category × Cohort Period. Right: Category × Academic Major Group.

Category × Cohort	Internships: 26.6% (2023-H1) → 12.2% (2024) — worsening trend. Competitions spike to 98.2% in 2024. Courses decline from 97.7% to 76.2%. Engagements stable at 95-97%.
Category × Major Group	Engagement opportunities show universally high predictions (95-99%) across all major groups. Internship predictions are lowest for Computing & IT (19.3%) and Unknown major (13.5%). Engineering students show the strongest internship probability at 32.4%.

Section 5: Feature Importance Analysis

Feature importance scores quantify the relative contribution of each input variable to the model's predictions. Both Random Forest (using mean decrease in impurity) and XGBoost (using gain-based importance) independently confirm the same top predictors — a strong validation signal.

5.1 Feature Importance Visualisation

The charts below display all 16 features ranked by importance score for both tree-based models. Features are colour-coded by importance tier: High (>0.10), Medium (0.05-0.10), Low (<0.05).

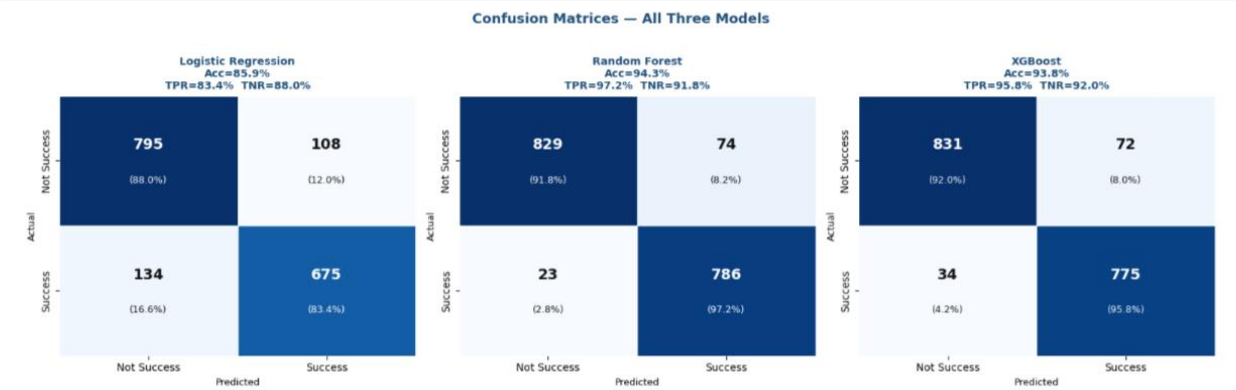

Figure 5: Feature Importance — Random Forest (left) and XGBoost (right). Green = High importance, Orange = Medium, Blue = Low.

5.2 Top Feature Importance Scores

Feature	RF Importance	XGB Importance	Tier
Duration Band	**0.582**	**0.505**	**HIGH**
Opportunity Category	**0.244**	**0.252**	**HIGH**
Apply Month	0.040	0.032	LOW
Apply Year	0.025	0.092	MEDIUM (XGB)
Signup Month	0.017	0.011	LOW
Cohort Period	0.015	0.023	LOW
Days to Apply	0.014	0.010	LOW
Signup Hour	0.012	0.008	LOW
Signup Year	0.011	0.005	LOW
Signup Day of Week	0.008	0.008	LOW
Academic Major	0.008	0.011	LOW
Country Group	0.007	0.009	LOW
Signup Quarter	0.007	0.014	LOW
Application Speed	0.005	0.008	LOW
Gender	0.003	0.007	LOW
Weekend Flag	0.002	0.007	LOW

5.3 Key Interpretations

Duration Band (RF: 0.582 | XGB: 0.505)

The single most powerful predictor in both models. This feature encodes the intensity and length of an opportunity and serves as a proxy for commitment level. Longer-duration opportunities attract a distinct applicant profile with significantly higher completion intent.

Opportunity Category (RF: 0.244 | XGB: 0.252)

The second-ranked feature. Category fundamentally determines success probability: Engagements and Courses predict near-certain participation; Internships predict systematic under-delivery (average 12–20%). This structural gap reflects supply-demand misalignment, not applicant quality.

Apply Year & Apply Month (XGB: 0.092 | 0.032)

Temporal features carry moderate-to-low importance. XGBoost elevates Apply Year to medium importance (0.092) — capturing the competitive shift across cohort years. Apply Month reflects intra-year seasonality in both models.

Actionable Low-Importance Features

While Country Group, Academic Major, and Application Speed score low overall (<0.011), their **segment-level effects are operationally significant**. For example, deliberate applicants (30–90 Days_to_Apply) outperform platform average by +7pp, and Engineering students show comparatively higher internship success rates.

Appendix

A. Technical Environment

Component	Details
Platform	Google Colab (GPU-enabled)
Language	Python 3.10
Core Libraries	scikit-learn 1.3, XGBoost 1.7, pandas 2.0, numpy 1.24, matplotlib 3.7, seaborn 0.12
Random Seed	42 (all models)
Notebook Link	https://colab.research.google.com/drive/1clXSNNUv3FiPZ1cAGI4MhaJHYseUPCGF
Dataset Source	RIT_Week2_Raw_Enriched_Dataset.csv — 8,558 rows, 43 columns post-enrichment
Target Variable	Success (binary: 1 = participated, 0 = did not)
Evaluation Metrics	AUC-ROC, F1-Score, Precision, Recall, Accuracy, 5-Fold CV

B. Model Reproducibility Notes

- All models use random_state=42 to ensure reproducible training splits and results.
- Stratified 80/20 split ensures class proportions are maintained in both train and test sets.
- StandardScaler is fit exclusively on training data; the same scaler is then applied to the test set to prevent data leakage.
- XGBoost scale_pos_weight is computed automatically as the ratio of the negative to positive class count in the training set.
- 5-fold CV is stratified and uses the same random seed. CV AUC reported as mean across all folds.

C. Categorical Encoding Reference

All categorical features were label-encoded (integer mapping) rather than one-hot encoded, to be compatible with tree-based model requirements and reduce dimensionality.

Column	Encoding Logic
Opportunity Category	LabelEncoder — 5 classes: Competition, Course, Engagement, Event, Internship
Academic Major Group	LabelEncoder — 7 groups including Business & Management, Engineering, Computing & IT, etc.
Country Group	LabelEncoder — Regional groupings: Africa, MENA, South Asia, Southeast Asia, etc.
Duration Band	Ordinal encoder: Short / Medium / Long / Extended mapped to integers
Application Speed	Ordinal: Same-day / 1-7 days / 8-30 days / 30-90 days / 90+ days
Cohort Period	Ordinal: 2022 / 2023-H1 / 2023-H2 / 2024 mapped to 0-3

D. Glossary

Term	Definition
AUC-ROC	Area Under the Receiver Operating Characteristic Curve — measures discriminative power. 0.97+ = Outstanding.
Precision	Of all predicted successes, what fraction were actually successful. Reduces false positives.
Recall (TPR)	Of all actual successes, what fraction were correctly identified. Reduces false negatives.
F1-Score	Harmonic mean of Precision and Recall — balances both trade-offs into one metric.
Ensemble	A weighted combination of multiple model predictions, reducing individual model bias and variance.
Stratified Split	Train/test split that preserves the class ratio in both subsets, ensuring fair

Term	Definition
	evaluation.
Feature Importance	A score indicating how much each feature contributes to prediction accuracy. Calculated differently in RF (impurity) vs XGBoost (gain).
Duration Band	Engineered feature grouping opportunity duration into intensity buckets (Short/Medium/Long/Extended).

E. Source References

- **Week 2 Enriched Dataset:** https://docs.google.com/spreadsheets/d/1wWGSWmO2uGUI4XRNtFf-BkeRO1gQWyAucmDuSwqM76A/edit?usp=sharing
- **Google Colab Notebook (Week 3):** https://colab.research.google.com/drive/1clXSNNUv3FiPZ1cAGI4MhaJHYseUPCGF?usp=sharing#scrollTo=IxKpHbdt0yJw
- **scikit-learn Documentation:** https://scikit-learn.org/stable/modules/generated/sklearn.ensemble.RandomForestClassifier.html
- **XGBoost Documentation:** https://xgboost.readthedocs.io/en/stable/

End of Report — RIT Data Insight Internship, Week 3 | May 2026